FLUID POWER SYSTEMS FOR INDUSTRIAL AUTOMATION

LAB MANUAL

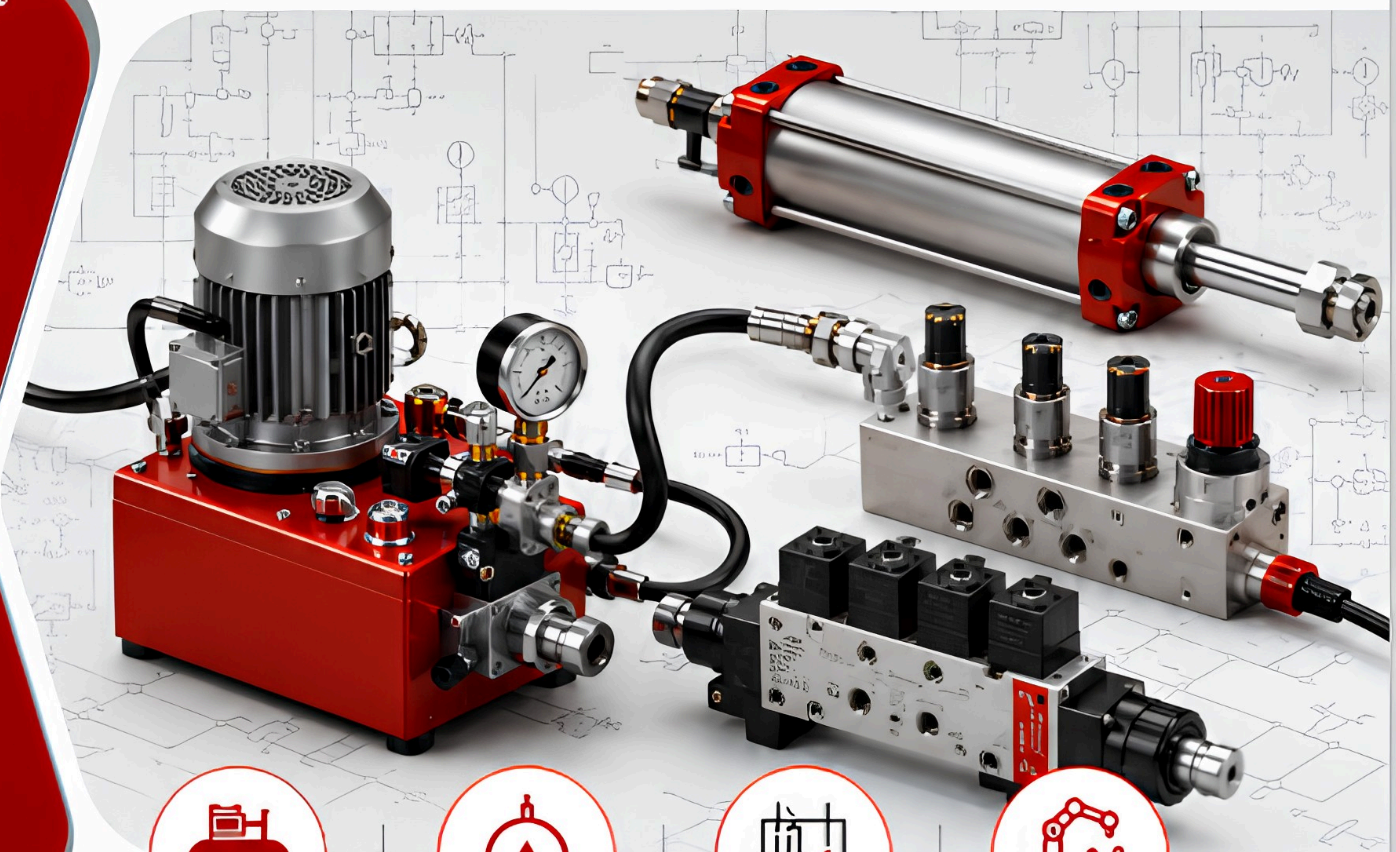
Experiments in Pneumatic, Hydraulic and Electro-Pneumatic Circuits

PNEUMATICS

HYDRAULICS

CONTROL ELEMENTS

ACTUATION & POWER



PNEUMATICS

HYDRAULICS

ELECTRO-PNEUMATICS

INDUSTRIAL APPLICATIONS

NAME : ....

ROLL NO. : ....

BRANCH : ....

SEMESTER : ....

AMRITA VISHWA VIDYAPEETHAM | School of Engineering
DEEMED TO BE UNIVERSITY

DEPARTMENT OF MECHANICAL ENGINEERING, COIMBATORE

23ARE303
FLUID POWER SYSTEMS FOR INDUSTRIAL AUTOMATION

LAB MANUAL

Certified that this is the bonafide record of work done by Mr./Ms. ………………………………………………………………………………………………………Register Number ……………………………………………… of ……………. Semester B.Tech……………………………………………………………………………… Branch in the………………………………………………………………………Laboratory/Workshop of this institute during the Academic year………………………………………

Faculty in-charge **Chairperson**

(For Internal Circulation Only)

Contents

General Laboratory Instructions

1. Come to the lab in uniform.
2. Enter the lab with Identity Card.
3. During the lab session, your first objective is to learn about the system, component or controller that you are working on.
4. Lab reports should be self-content and neatly written. They should contain your circuit diagrams as well as an explanation.
5. The exercises on the laboratory record are base line exercises. You are encouraged to formulate additional questions and to test them using the lab set up. How much you learn depends on how willing you are to ask additional questions.
6. Expect to do most of your learning in the lab or in front of your computer, rather than from lectures.
7. Your lab report and your modeling exercises should be your own work and reflect your understanding of the problem. No copying allowed, you are encouraged to work together as a group.
8. Fair records are to be submitted within one week of doing the exercise.
9. Listen to the instructions carefully.
10. Design your circuit before the lab session.
11. Follow the lab manual / record; don't create your own symbols.
12. Non-labelled diagrams are considered wrong diagrams.
13. Don't start constructing your design before you have all the instructions.
14. Unplug the power source (Pneumatic or electrical) while constructing your design.
15. Demonstrate your working design.
16. After simulating the circuit remove the components and put them in place.
17. Lab groups will be composed of minimum three students, so carryout the simulation with the team spirit and group learning.

INTRODUCTION TO FLUID POWER SYSTEMS

Fluid Power

Fluid power is the technology in which energy is transmitted and controlled using a pressurized fluid. The fluid may be either a liquid or a gas. Based on the working medium used, fluid power systems are mainly classified into two types: pneumatic systems and hydraulic systems.

In pneumatic systems, compressed air or other neutral gases are used to transmit and control power. In hydraulic systems, pressurized liquid, usually oil, is used as the working medium. Both systems are widely used in industries because they can transmit power efficiently, produce useful mechanical motion, and provide controlled operation of machines.

In many machines, power must be transmitted from one location to another. This can be achieved mechanically, electrically, hydraulically, or pneumatically. In fluid power systems, the pressurized fluid carries energy to actuators, which convert it into linear or rotary motion. Fluid power systems are commonly used in automated machinery, clamping devices, material handling systems, vehicle braking systems, presses, construction equipment, and excavators. These systems can also be combined with sensors, switches, solenoid valves, relays, microprocessors, and PLCs to develop automated control systems.

Advantages of Fluid Power Systems

Fluid power systems offer several important advantages:

- They can multiply and vary force easily
- They provide smooth and accurate control of motion
- They allow multiple operations to be controlled from one power source
- They have a good power-to-weight ratio
- They can produce controlled force and torque
- They are suitable for many industrial automation applications
- Standard components are widely available, making design and maintenance easier

PNEUMATIC SYSTEM

Introduction to Pneumatics

Pneumatics deals with the use of compressed air to transmit and control power. The word pneumatics is derived from the Greek word pneuma, which means breath or wind. Since air is freely available in the atmosphere, pneumatic systems are widely used in industrial automation.

In pneumatic systems, atmospheric air is compressed, stored, treated, and supplied to different components such as valves and actuators. The compressed air performs useful work by producing motion in cylinders or rotary actuators.

Pneumatic systems are commonly used for simple and fast operations such as clamping, shifting, feeding, sorting, positioning, stamping, packaging, and material handling. They are also used in automated machines where quick response and clean operation are required.

Applications of Pneumatic Systems

Pneumatic systems are widely used in the following applications:

- Clamping
- Shifting
- Positioning
- Orienting
- Packaging
- Feeding
- Metering
- Door and chute control
- Transfer of materials
- Sorting of parts
- Stacking of components
- Stamping and embossing
- Drilling, turning, milling, sawing, and finishing operations

Pneumatic components can produce different types of motion such as:

- Linear motion
- Rotary motion

- Swivelling motion

Basic Components of a Pneumatic System

A pneumatic system generally consists of the following components:

- Compressor
- Receiver
- Filter
- Dryer
- Lubricator
- Pressure regulator
- Directional control valves
- Flow control valves
- Pneumatic cylinders
- Sensors and switches
- Connecting pipes and fittings

The compressor takes air at atmospheric pressure and compresses it to a higher pressure. The receiver stores the compressed air and helps maintain constant pressure. The filter removes dust and condensate from the air. The dryer removes moisture. The pressure regulator maintains the required working pressure. Valves control the direction and flow of air, while actuators convert compressed air energy into mechanical motion.

Pressure in Pneumatic Systems

Pressure is force acting at perpendicular to a surface area. 1 Pascal represents a constant pressure on a surface area of 1 sq. m. with a force of 1N acting at right angles to that surface area. 100 KPa is equal to 1 bar, and this represents atmospheric air pressure. As can be seen, the Pascal is a small unit of pressure and hence we tend to use the bar. One (1) bar represents atmospheric pressure. Pressure is defined as force F per unit area A.

$$P = F/A \ (N/m^2)$$

The unit of pressure is called Pascal

$$1 \text{ Pa} = 1 \text{ N/m}$$

$$100{,}000 \text{ Pa} = 1 \text{ bar}$$

$$100{,}000 \text{ N/m}^2 = 1 \text{ bar (or) } 1 \text{ bar} = 10 \text{ N/cm}^2$$

Normally, pneumatic systems operate at a working pressure of about 5 to 6 bar.

Advantages of Pneumatic Systems

Pneumatic systems have several advantages:

- Air is freely available in the atmosphere
- Compressed air can be easily transported through pipelines
- Air can be stored in receivers and used when required
- Pneumatic systems are suitable for high-speed operation
- Components are simple and relatively inexpensive
- The system is clean and safe for many industrial applications
- Pneumatic tools and actuators are overload safe
- Compressed air is less affected by temperature changes
- Pneumatic systems are suitable for fire-prone and explosive environments

Disadvantages of Pneumatic Systems

The main limitations of pneumatic systems are:

- Compressed air requires proper preparation
- Air must be filtered and dried before use
- Since air is compressible, uniform speed control is difficult
- Pneumatic systems are not suitable for very high force applications
- Exhaust air may produce noise
- Generation of compressed air can be costly
- Leakage reduces efficiency and increases energy loss

Basic Requirements of Compressed Air Supply

For effective operation of pneumatic systems, the compressed air must be:

- Clean
- Dry
- Free from unwanted oil and moisture

Air preparation is very important because contamination can damage pneumatic components and reduce system performance. Filters, dryers, lubricators, regulators, drain valves, and oil separators are used to improve the quality of compressed air.

Electro-Pneumatics

Electro-pneumatics is the combination of electrical control and pneumatic power. In this system, electrical signals are used to control pneumatic valves and actuators. This makes the system faster, more flexible, and suitable for automation.

In an electro-pneumatic system, sensors or switches detect the condition of the machine. The signal is then processed using relays, timers, controllers, or PLCs. The output signal energizes a solenoid valve, which controls the flow of compressed air to the actuator.

The main elements of an electro-pneumatic system are:

- Pneumatic cylinders
- Rotary actuators
- Solenoid-operated directional control valves
- Relays
- Push buttons
- Limit switches
- Proximity sensors
- Compressor
- Receiver

Electro-pneumatic systems are widely used in automated machines because they provide quick response, easy control, and flexibility. They are especially useful when the operating sequence needs to be changed according to production requirements.

Signal Flow in Pneumatic Systems

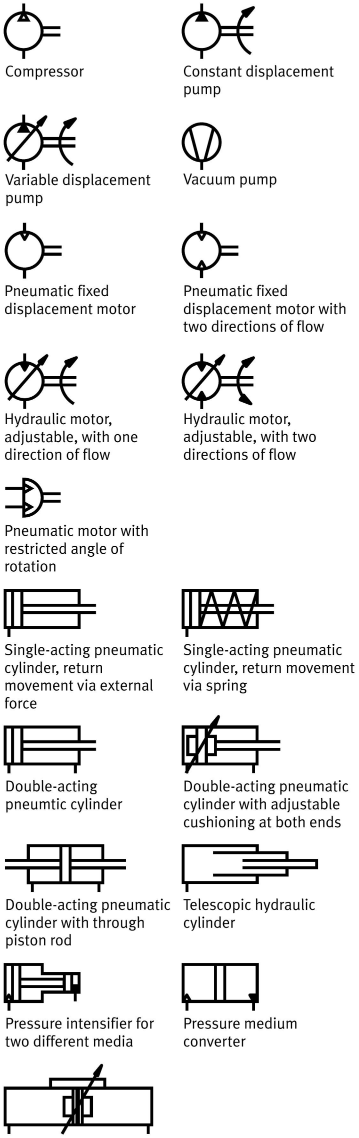
A pneumatic control system consists of different groups of elements connected together to control the movement of actuators. The signal flow normally starts from the input section and ends at the actuator.

The basic levels of signal flow are:

1. Energy supply
2. Input elements or sensors
3. Processing elements
4. Control elements
5. Power components or actuators

Input elements receive information from the machine or operator. Processing elements decide the required action. Control elements direct the flow of compressed air. Finally, actuators convert compressed air energy into mechanical motion.

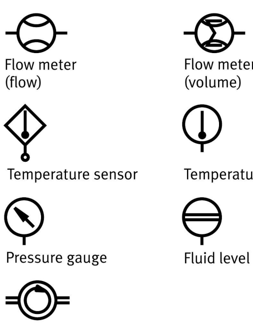
Energy conversion



Compressor
Constant displacement pump
Variable displacement pump
Vacuum pump
Pneumatic fixed displacement motor
Pneumatic fixed displacement motor with two directions of flow
Hydraulic motor, adjustable, with one direction of flow
Hydraulic motor, adjustable, with two directions of flow
Pneumatic motor with restricted angle of rotation
Single-acting pneumatic cylinder, return movement via external force
Single-acting pneumatic cylinder, return movement via spring
Double-acting pneumtic cylinder
Double-acting pneumatic cylinder with adjustable cushioning at both ends
Double-acting pneumatic cylinder with through piston rod
Telescopic hydraulic cylinder
Pressure intensifier for two different media
Pressure medium converter
Pneumatic linear unit

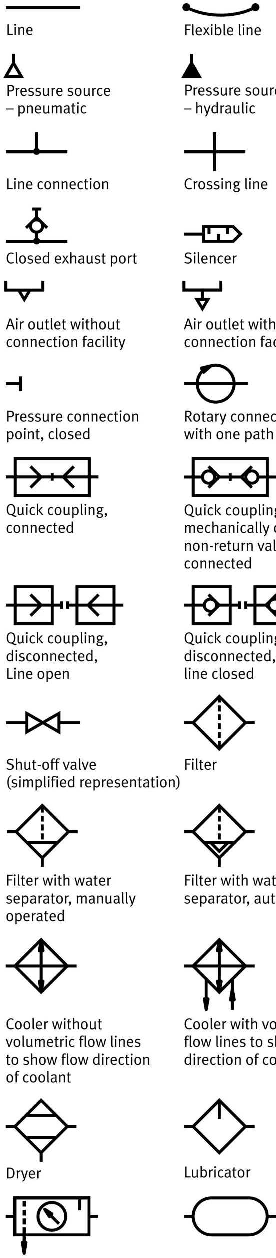
Other symbols

Other devices



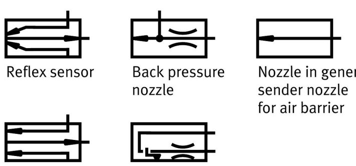
Flow meter (flow)
Flow meter (volume)
Temperature sensor
Temperature gauge
Pressure gauge
Fluid level gauge
Tachometer

Transfer of ener gy



Line
Flexible line
Pressure source – pneumatic
Pressure source – hydraulic
Line connection
Crossing line
Closed exhaust port
Silencer
Air outlet without connection facility
Air outlet with connection facility
Pressure connection point, closed
Rotary connection with one path
Quick coupling, connected
Quick coupling with mechanically opening non-return valves, connected
Quick coupling, disconnected, Line open
Quick coupling, disconnected, line closed
Shut-off valve (simplified representation)
Filter
Filter with water separator, manually operated
Filter with water separator, automatic
Cooler without volumetric flow lines to show flow direction of coolant
Cooler with volumetric flow lines to show flow direction of coolant
Dryer
Lubricator
Service unit (simplified representation)
Reservoir

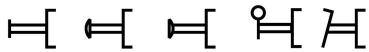
Special symbols



Reflex sensor
Back pressure nozzle
Nozzle in general, sender nozzle for air barrier
Receiver nozzle (with air supply) for air barrier
Pneumatic proximity switch, actuated by permanent magnet

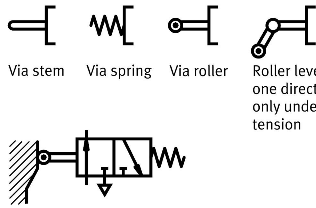
Actuation methods

Manual operation



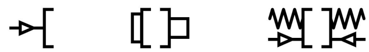
General
Via push button
Via pull
Via lever
Via pedal

Mechanical actuation



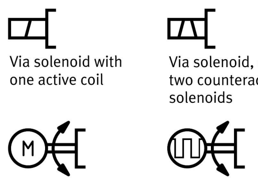
Via stem
Via spring
Via roller
Roller lever in one direction only under tension
Actuated in initial position

Pressure actuation



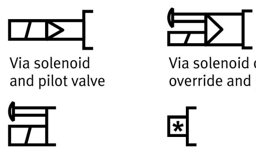
Direct via pneumatic actuation
Via differential pressure actuation
Spring centered

Electrical actuation



Via solenoid with one active coil
Via solenoid, e. g. with two counteracting solenoids
Via electric motor with continuous rotation
Via electric stepper motor

Combined actuation



Via solenoid and pilot valve
Via solenoid or manual override and pilot valve
Via solenoid or manual operation
General explanatory symbol

Mechanical actuating components

Detent
Over-centre device

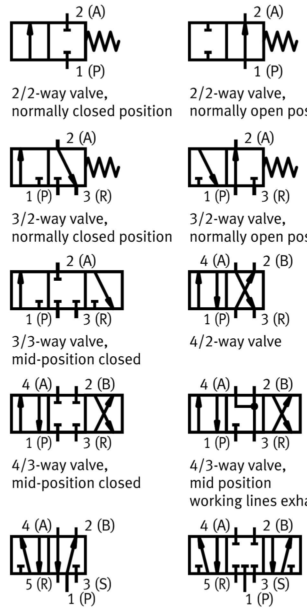
Connection designations

ISO/ DIS 11727	for 2/2- and 3/2-way valves	for 4/2- and 4/3-way valves	for 5/2- and 5/3-way valves
1	P	P	P
2	A	B	B
3	R	R	S
4	–	A	A
5	–	–	R
10*	Z	–	–
12	Z	Y	Y
14	–	Z	Z

* Control port, on valves with normally open position (NOT function)

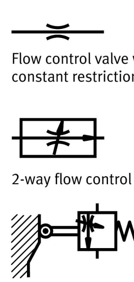
Valves

Directional control valves



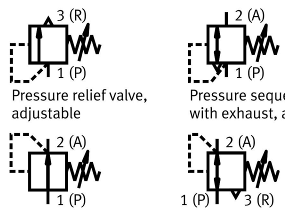
2/2-way valve, normally closed position
2/2-way valve, normally open position
3/2-way valve, normally closed position
3/2-way valve, normally open position
3/3-way valve, mid-position closed
4/2-way valve
4/3-way valve, mid-position closed
4/3-way valve, mid position working lines exhausted
5/2-way valve
5/3-way valve, mid-position closed
Directional control valve with intermediate switching positions and two end positions

Flow control valve



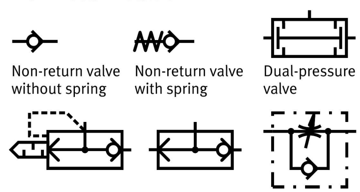
Flow control valve with constant restriction
Flow control valve, adjustable, any actuation
2-way flow control valve
Flow divider
Flow control valve, adjustable, mechanical actuation against spring

Pressure valve



Pressure relief valve, adjustable
Pressure sequence valve with exhaust, adjustable
Pressure regulating valve, non-relieving, adjustable
Pressure regulating valve, relieving type, adjustable

Non-return valve



Non-return valve without spring
Non-return valve with spring
Dual-pressure valve
Quick exhaust valve
Shuttle valve
One-way flow control valve, adjustable

HYDRAULIC SYSTEM

Introduction to Hydraulics

A hydraulic system uses pressurized liquid to transmit and control power. The working fluid is usually hydraulic oil. Hydraulic systems are used when high force, smooth motion, and accurate control are required.

The main function of a hydraulic system is to convert mechanical energy into hydraulic energy and then convert it back into mechanical energy using an actuator.

Components of a Hydraulic System

The power supply section of a hydraulic system generally consists of the following components:

- Drive unit
- Pump
- Pressure relief valve
- Coupling
- Reservoir
- Filter
- Cooler
- Heater
- Connecting pipes and hoses
- Safety and monitoring devices

The drive unit, usually an electric motor or an engine, provides mechanical power to the pump. The pump draws hydraulic fluid from the reservoir and delivers it into the system. As the fluid flows through pipes, valves, and actuators, resistance is developed. This resistance causes pressure to build up in the system.

A very important point is that the pump does not directly create pressure. The pump creates flow. Pressure is developed only when there is resistance to the flow. If the outlet of a positive displacement pump is blocked, pressure can increase rapidly and damage the system. To avoid this, a pressure relief valve is used to limit the maximum system pressure.

Hydraulic Pumps

Hydraulic pumps convert mechanical energy into hydraulic energy. Pumps used in hydraulic systems are mainly classified into two types:

1. Non-positive displacement pumps

Non-positive displacement pumps produce continuous flow, but their output changes when pressure changes. Centrifugal and propeller pumps are examples of non-positive displacement pumps. These pumps are not commonly used in power hydraulic systems.

2. Positive displacement pumps

Positive displacement pumps deliver a fixed quantity of fluid during each cycle of operation. Their flow rate remains almost constant even when pressure changes. These pumps are commonly used in hydraulic power systems because they can develop high pressure.

Common types of positive displacement pumps include:

- Gear pumps
- Vane pumps
- Piston pumps
- Gerotor pumps

Piston pumps are generally preferred for high-pressure applications. Gear pumps are simple, compact, and widely used in many hydraulic systems.

Hydraulic Actuators

A hydraulic actuator converts hydraulic energy into mechanical force and motion. Hydraulic actuators are mainly classified into two types:

1. Linear actuators

Linear actuators produce motion in a straight line. Hydraulic cylinders are the most common example. They are also known as rams or reciprocating motors.

Hydraulic cylinders may be:

- Single-acting cylinders
- Double-acting cylinders

In a single-acting cylinder, hydraulic pressure acts only in one direction, and the return stroke is usually achieved by a spring or external load. In a double-acting cylinder, hydraulic pressure can act on both sides of the piston, allowing controlled extension and retraction.

2. Rotary actuators

Rotary actuators produce rotary motion and torque. Hydraulic motors are examples of rotary actuators. They convert hydraulic energy into mechanical rotation and are used where continuous rotary motion is required.

Hydraulic Motors

A hydraulic motor converts hydraulic energy into mechanical energy in the form of rotary motion. In a hydraulic transmission system, the pump supplies pressurized fluid to the motor. The motor then converts this fluid power into torque and rotation to drive a machine or load.

The main ratings of a hydraulic motor are:

- Torque
- Pressure
- Displacement
- Speed

Hydraulic motors may be fixed displacement or variable displacement. Common types of hydraulic motors include gear motors, vane motors, and piston motors.

Hydraulic Valves

Hydraulic valves are used to control the direction, pressure, and flow rate of hydraulic fluid. They play an important role in controlling the motion, speed, and force of actuators.

The three main types of hydraulic control valves are:

1. Pressure control valves

Pressure control valves protect the system from excessive pressure. They are also used for sequencing, unloading, counterbalancing, and pressure reduction.

Common pressure control valves include:

- Relief valves
- Safety valves
- Sequence valves
- Counterbalance valves
- Brake valves
- Unloading valves
- Pressure reducing valves

A relief valve is one of the most important safety components in a hydraulic system. It limits the maximum system pressure and prevents damage to components.

2. Directional control valves

Directional control valves control the path of fluid flow. They are used to start, stop, extend, retract, or reverse the movement of actuators.

Three-way directional control valves are commonly used with single-acting cylinders. Four-way directional control valves are commonly used with double-acting cylinders.

3. Flow control valves

Flow control valves control the speed of actuators by regulating the quantity of fluid flowing through the system. A simple orifice or needle valve can be used as a basic flow control device. Flow control valves may be:

- Fixed or adjustable
- Compensated or non-compensated
- Restrictor type or bypass type

Pressure-compensated flow control valves maintain nearly constant flow even when load conditions change.

Limitations of Hydraulic Systems

Although hydraulic systems are powerful and reliable, they have some limitations:

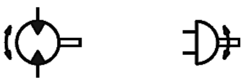
- Leakage may occur at seals, joints, and connections.
- Hydraulic power requires a pump, which may be heavy and noisy.
- Hydraulic fluid may get contaminated and therefore requires proper filtration.
- Air trapped in the system can cause spongy operation and loss of accuracy.
- Cavitation may occur if the pump inlet pressure is too low.
- The design and modelling of hydraulic systems can be more complex than simple electrical systems.

Energy conversion


hydraulic pump with constant delivery rate

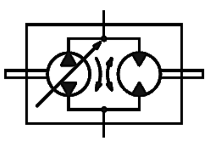

hydraulic variable displacement pump with alternating direction of delivery at constant direction of rotation

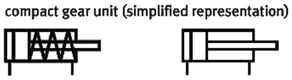

reversible hydraulic pump/hydraulic motor unit with two directions of flow and variable displacement volume, external leakage oil line and two directions of rotation.


hydraulic motor with constant displacement/ absorption capacity and two directions of rotation

rotary drive/semi-rotary drive with limited angle of rotation and two directions of flow

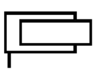

electric motor

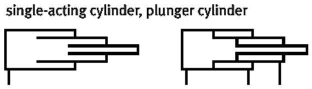

compact gear unit (simplified representation)


single-acting cylinder with piston rod at one end, spring chamber with leakage oil connection

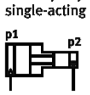
double-acting cylinder with piston rod at one end


double-acting cylinder with through piston rod of different diameters, with adjustable cushioning at both ends

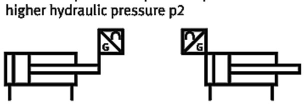

single-acting cylinder, plunger cylinder


telescope cylinder, single-acting

telescope cylinder, double-acting



pressure booster, single-acting, which converts a pneumatic pressure p1 into a higher hydraulic pressure p2


double-acting cylinder with displacement encoder on the piston rod

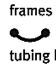
double-acting cylinder with integrated displacement encoder

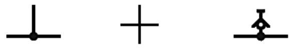
Energy transfer

line, supply line, return line, component framing and symbol boxed in

internal and external pilot line, leakage oil line, purging line, exhaust line


frames for several components


tubing line


line connection

crossing line

closed exhaust port

pressure supply port, closed

plug within a fluid line


hydraulic energy source

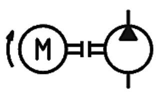
tank connection, ending below and above the oil level


reservoir closed with cover

reservoir open to atmosphere


motor shaft points to the right (preferably) or to the left

pump shaft points to the left (preferably) or to the right

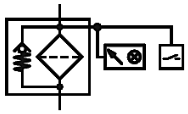

hydraulic pump and hydraulic motor with shaft coupling

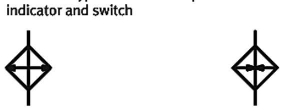

quick decoupler with two non-return valves, decoupled; i.e. line closed

quick decoupler with two non-return valves, coupled

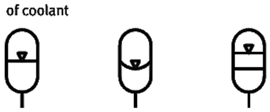

filter

tank – ventilation filter


filter with bypass valve with optical contamination indicator and switch


cooler without flow lines for the direction of flow of coolant

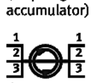
heating


gas pressure accumulator, separation of media through:

diaphragm (diaphragm accumulator)

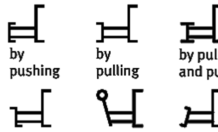
bladder (bladder accumulator)

piston (piston accumulator)


three-way rotary connection

Actuation

Manual actuation


by pushing

by pulling

by pulling and pushing

by turning

by lever

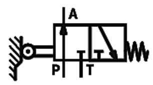
by rocker

Mechanical actuation


by plunger

by roller

by roller lever, actuation in one direction of travel only

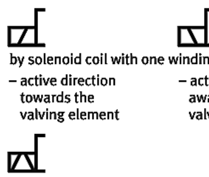

actuated in initial position

Pressure actuation


pneumatically actuated

hydraulically actuated

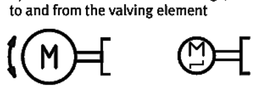
Electrical actuation


by solenoid coil with one winding

– active direction towards the valving element

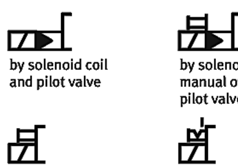
– active direction away from the valving element

by solenoid coil with two windings, active direction to and from the valving element


by electric motor with continuous rotary motion

by stepper motor

Combined actuation


by solenoid coil and pilot valve

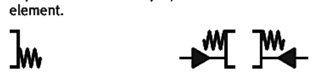
by solenoid coil or manual override and pilot valve

by solenoid coil or manual override

by solenoid coil or manual override with detent

Mechanical components

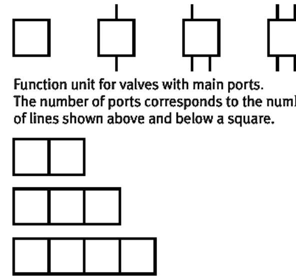

Detents are to be symmetrically matched with the mechanical connection.
With more than three detent positions the number of positions can be displayed above the detent element.


reset by spring

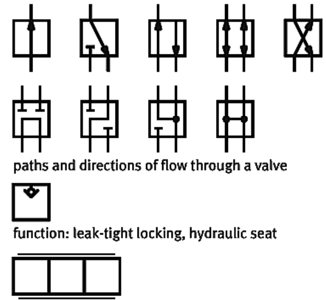
spring-centred

Valves

Basic symbols

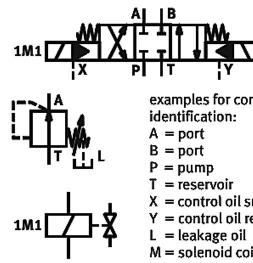

Function unit for valves with main ports.
The number of ports corresponds to the number of lines shown above and below a square.

basic elements with 2, 3, 4 switching positions


paths and directions of flow through a valve

function: leak-tight locking, hydraulic seat

directional control valve with intermediate positions – dynamic control valve



examples for connection identification:
A = port
B = port
P = pump
T = reservoir
X = control oil supply
Y = control oil relief
L = leakage oil
M = solenoid coil

Connections must be marked on the circuit diagram using the identification marked on the component, sub-base or block (ISO 1219-2).

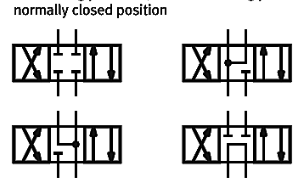

2/2-way valve with 2 ports, 2 switching positions for 2 flow directions.

– normally closed position

– normally open position


3/2-way valve with 3 ports and 2 switching positions, normally closed position

4/2-way valve with 4 ports and 2 switching positions


4/3-way valve with 4 ports and 3 switching positions – various mid-position variants

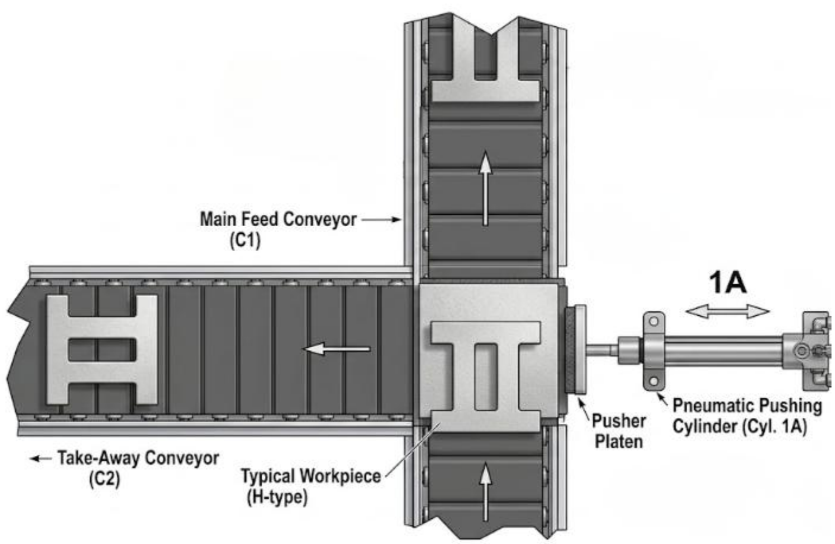
EXERCISE 1

Design Simulation and Testing of Pneumatic Circuits Using Single Acting and Double Acting Cylinder

Exp. No.: 1A	Design and Simulation of Circuits Using Single Acting Cylinder for Specific Applications
Date:	

PROBLEM:

Through operation of the push button on the actuating valve, metal stampings lying in random positions are sorted out and transformed to a second conveyor belt. The forward motion of the single acting cylinder is to be adjusted. When the push button is released, the piston rod travels to the retracted end position.



WORKING PRINCIPLE:

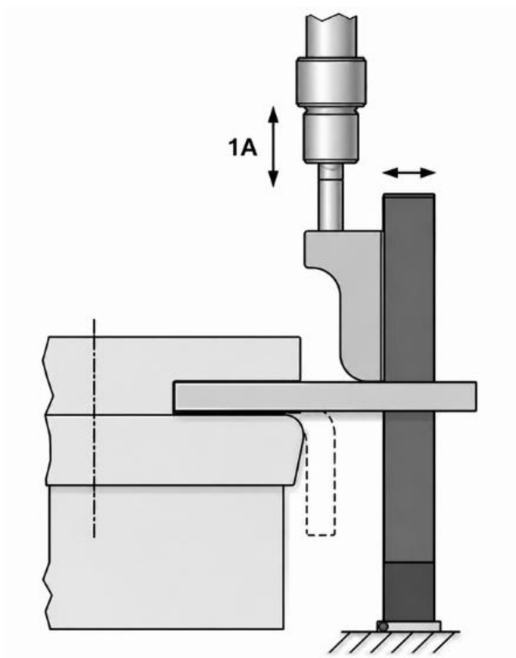
PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 1B	Design and simulation of circuits using double acting cylinder for specific applications
Date:	

PROBLEM:

Operation of two identical valves by push button causes the forming tool of an edge folding device to thrust downwards and fold over the edge of a flat sheet of cross-sectional area 40 x 5. If both or even just one push button is released, double acting cylinder (1A) slowly returns to the initial position.



WORKING PRINCIPLE:

PNEUMATIC CIRCUIT:

INFERENCE:

EXERCISE 2

Design, simulation and testing of pneumatic circuit using logic elements, flow control valves and quick exhaust valves

Exp. No.: 2A	**Design and simulation of a clamping of workpiece and drilling system**
Date:	

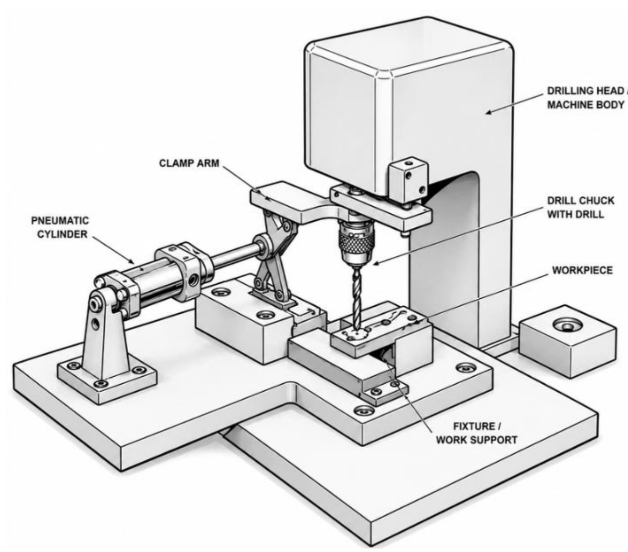
PROBLEM:

Clamping of a work piece:

A flat rectangular plate has to be drilled. Clamping of the plate is done by a double acting cylinder subject to the availability of work piece. The clamping of the plate must be possible by manual control either by pressing a push button or by a foot pedal. After the plate is firmly clamped drilling operation is to be performed. Once the drilling operation is over then only the work piece is unclamped. Unclamping must be initiated by a further manual pushbutton.

Conditions:

a) Clamping must be possible only when the workpiece has been inserted.
b) Unclamping must not be possible during the drilling operation.
c) It must be possible to set the clamping operation (advance stroke) to a slow speed.
d) Unclamping (return stroke) must be carried out very quickly.



WORKING PRINCIPLE:

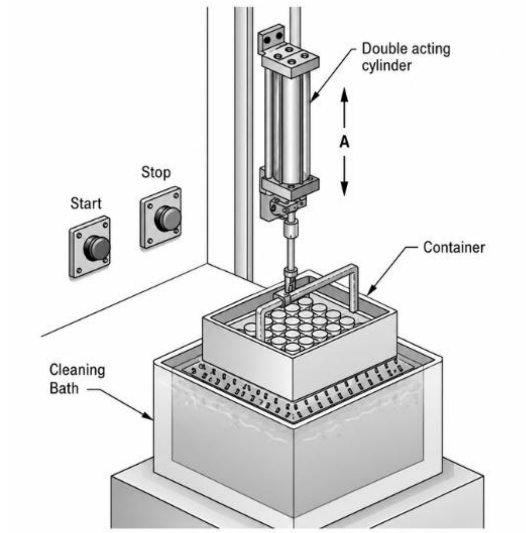
PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 2B	Design and simulation of a cleaning bath system
Date:	

PROBLEM:

A container is to be moved up & down in the cleaning bath continuously. The "START" and "STOP" signals for the continuous cycle are to be given by a pushbutton. The effect of the "STOP" signal should be such that the container stops in the raised position. A double acting cylinder is to be controlled such that, after being given the start signal, the piston performs oscillatory motion between the end positions until the reverse signal is input. The cylinder should then remain stationary in the rear end position.



WORKING PRINCIPLE:

PNEUMATIC CIRCUIT:

INFERENCE:

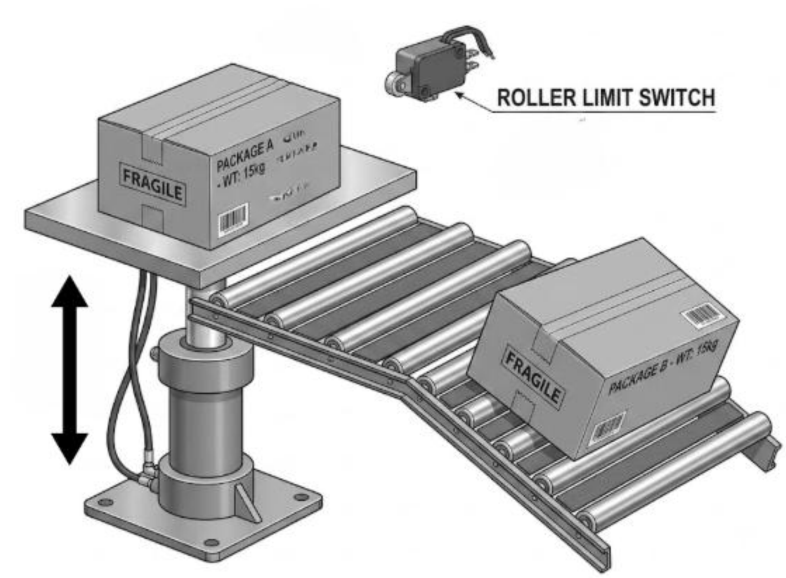
EXERCISE 3

Design, simulation and testing of multiple actuator pneumatic circuits using cascade method for various industrial applications

Exp. No.: 3A	**Design and simulation of pneumatic lifting system with sensing valve**
Date:	

PROBLEM:

Boxes are lifted to conveyor height by a double acting cylinder. The piston rod is to extend fully when a switch is operated. The piston rod is to extended position before the operator can initiate rod retraction by a second switch. A roller limit switch confirms full extension. The rod is to continue forward even if the extension switch is released before full extension is reached.



WORKING PRINCIPLE:

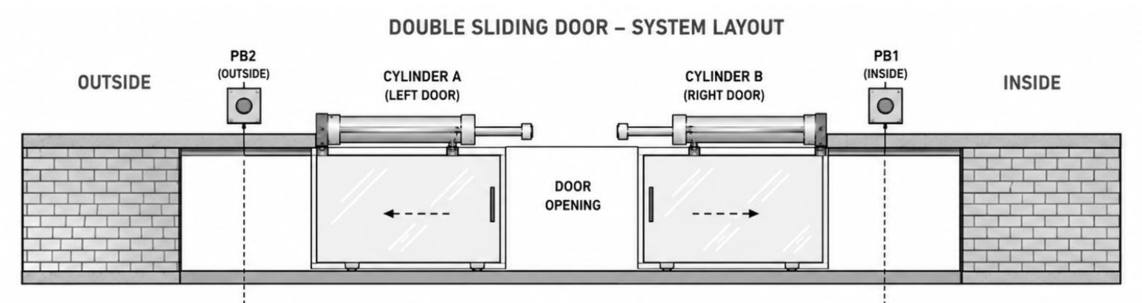
PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 3B	Automatic Open and Door Closing System Using Pneumatic Cylinder
Date:	

PROBLEM:

A pneumatically controlled double sliding door is to be used in a crockery mart. Double sliding door is to be controlled either from outside or from inside by pressing a pushbutton. The opening and closing operations are initiated by the same pushbutton in each case. The speeds of the double sliding door movements are to be adjustable. Develop a pneumatic control circuit for the given task.



WORKING PRINCIPLE:

PNEUMATIC CIRCUIT:

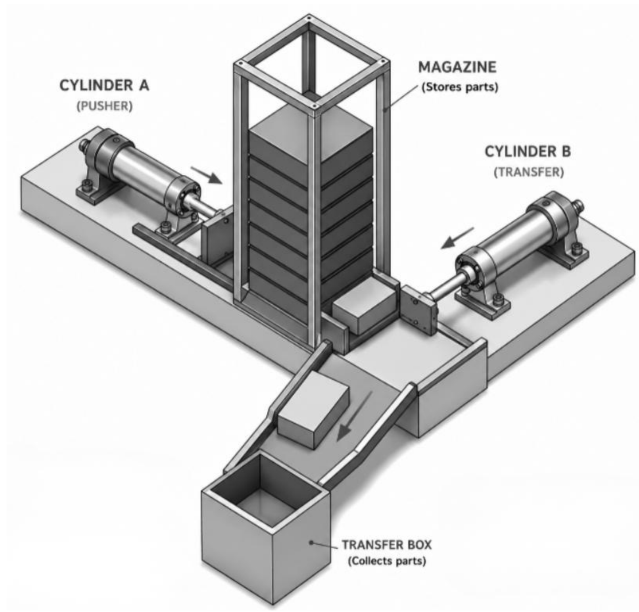
INFERENCE:

Exp. No.: 3C	**Pneumatic circuit design for a part transfer station**
Date:	

PROBLEM:

Two cylinders are used to transfer parts from a magazine onto a chute. When a pushbutton is pressed, cylinder (A) extends, pushing the part from the magazine and positions it in preparation for transfer by cylinder (B) onto the out-feed chute. Once the part is transferred, the cylinder (A) retracts, followed by the cylinder (B).

Sequence code: A1 B1 A0 B0



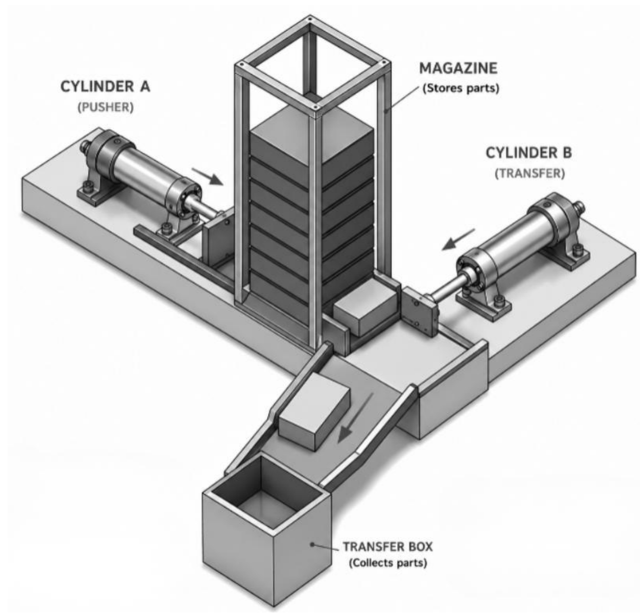
WORKING PRINCIPLE:

PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 3D	Pneumatic circuit design for a part transfer station using cascade method
Date:	

PROBLEM:



Two cylinders are used to transfer parts from a magazine onto a chute. When a pushbutton is pressed, cylinder (A) extends, pushing the part from the magazine and positions it in preparation for transfer by cylinder (B) onto the out-feed chute. Once the part is transferred, the cylinder (B) retracts, followed by the cylinder (A).

Sequence code: A1 B1 B0 A0

WORKING PRINCIPLE:

PNEUMATIC CIRCUIT:

INFERENCE:

Additional Questions / Notes:

Additional Questions / Notes:

Additional Questions / Notes:

EXERCISE 4

Design and simulation of hydraulic circuits circuit using single acting and double acting cylinder

Exp. No.: 4A	**Actuation of single acting hydraulic cylinder**
Date:	

PROBLEM:

A single acting hydraulic cylinder is to be extended when a lever type 3/2 spring return hydraulic control valve is actuated. Until the lever is in operating condition, the single actuating cylinder should be in extended position and after the lever is back to original position the cylinder should retract to the home position. Use a unidirectional fixed displacement pump along with a pressure relief valve.

WORKING PRINCIPLE:

HYDRAULIC CIRCUIT:

INFERENCE:

Exp. No.: 4B	Actuation of Double Acting Hydraulic Cylinder
Date:	

PROBLEM:

Design a hydraulic double-acting cylinder circuit using a hydraulic valve with a configuration of 4/2 lever operated spring return valve. The cylinder is to extend when a lever is operated and when the lever is not operated due to spring the cylinder is to retract. Design a hydraulic double-acting cylinder circuit using a hydraulic valve with a configuration of 4/3-way spring centred valve. The cylinder is to extend and return by actuating a separate lever for forward and return stroke.

WORKING PRINCIPLE:

HYDRAULIC CIRCUIT:

INFERENCE:

EXERCISE 5

Design and Simulation of Hydraulic Circuits Using Sequence Valve and Counterbalance Valve Applications

Exp. No.: 5A	**Actuation of Two Cylinders with the Help of a Pressure Sequence Valve**
Date:	

PROBLEM:

Design a two-cylinder Hydraulic sequencing circuit using two pressure sequence valves which allow the automatic sequencing of two cylinders.

- The actuator-1 moves as soon as fluid flow is directed to the piston end of the actuator.
- Sequence valve blocks flow to the actuator-2 until a predetermined pressure is reached after the pressure sequence valve allows fluid flow to the actuator-2.
- The pre-determined pressure will be developed after the complete extension of cylinder-1.
- A sequence valve is typically fitted with an integral check valve. This allows free flow of fluid around the valve when the direction of the actuator is reversed.
- After the full extension of cylinder-2 the oil goes through another pressure sequence valve after the pre-determined pressure to move the cylinder-1 to the home position followed by the cylinder-2.

Use a hydraulic valve with a configuration of 4/3 way valve with spring centred. The valve is operated from both sides by a lever.

WORKING PRINCIPLE:

HYDRAULIC CIRCUIT:

INFERENCE:

Exp. No.: 5B	Control Of Hydraulic Cylinder with the Help of a Counterbalance Valve
Date:	

PROBLEM:

A hydraulic cylinder is vertically mounted on which a load was mounted. Due to the applied load, cylinder may tend to extend automatically. If the load falls due to the extension of the cylinder automatically due to its self-weight, it leads to accident. Design a suitable hydraulic circuit to avoid the free falling of the hydraulic cylinder using a counterbalance valve with check valve. A 4/3-way valve spring centred valve can be used. Both sides of the valve are operated by a lever.

WORKING PRINCIPLE:

HYDRAULIC CIRCUIT:

INFERENCE:

EXERCISE 6

Design and Simulation of Hydraulic Circuit for Industrial Application

Exp. No.: 6A	Regenerative Circuit of a Double Acting Hydraulic Cylinder
Date:	

PROBLEM:

Design a regenerative-cylinder-advance circuit, which provides a rapid advance to the point where a load is encountered. Fluid recirculates from the rod end into the piston end and coupled with pump output to increase flow into the piston end, thus increasing cylinder extension speed. A 4/3 hydraulic control valve with the central position output and input is closed can be used. Another output is in blocked condition. A lever operates both sides of the valve. Drain port of the valve is directly connected to the tank.

WORKING PRINCIPLE:

HYDRAULIC CIRCUIT:

INFERENCE:

Exp. No.: 6B	**Design of multiple actuator hydraulic circuit using limit sensing valves.**
Date:	

PROBLEM:

Two double acting hydraulic cylinder is to be operated with a sequence of A1 B1 A0 B0.

A1 – Represents the forward stroke of the hydraulic cylinder "A"

B1 – Represents the forward stroke of the hydraulic cylinder "B"

A0 – Represents the Return stroke of the hydraulic cylinder "A" B0 – Represents the Return stroke of the hydraulic cylinder "B"

Use two 3/2 roller operated spring return hydraulic valve to sense the position of each hydraulic cylinder. A 3/2 lever operated spring return hydraulic valve should be used to start the sequence code. Ensure before starting the previous cycle was completed. Employ two 4/2 double pilot operated hydraulic valve to supply the oil to both cylinders.

WORKING PRINCIPLE:

HYDRAULIC CIRCUIT:

INFERENCE:

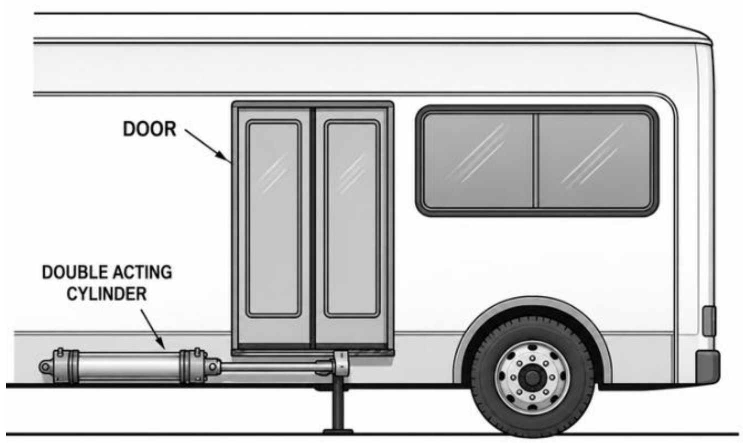
EXERCISE 7

Design and Simulation of Basic Electro-Pneumatic Circuits

<table>
<tr><td>Exp. No.: 7A</td><td rowspan="2">Automatic Door Opening and Closing of a Bus with Passenger Sensing</td></tr>
<tr><td>Date:</td></tr>
</table>

PROBLEM:

The automatic door of a bus is operated by a Bus driver. When the bus driver presses an electrical push button a double acting cylinder extends and opens the door. The passenger entry is monitored by a sensor. This sensor output should be memorized using a relay and its own contact. The driver presses another electrical push button after the confirmation of passenger entry to close the door. The relay should be switched off using the output of door closing limit switch. Both the door opening speed and closing speed is to be adjustable separately. Design and draw an Electro-pneumatic circuit.



WORKING PRINCIPLE:

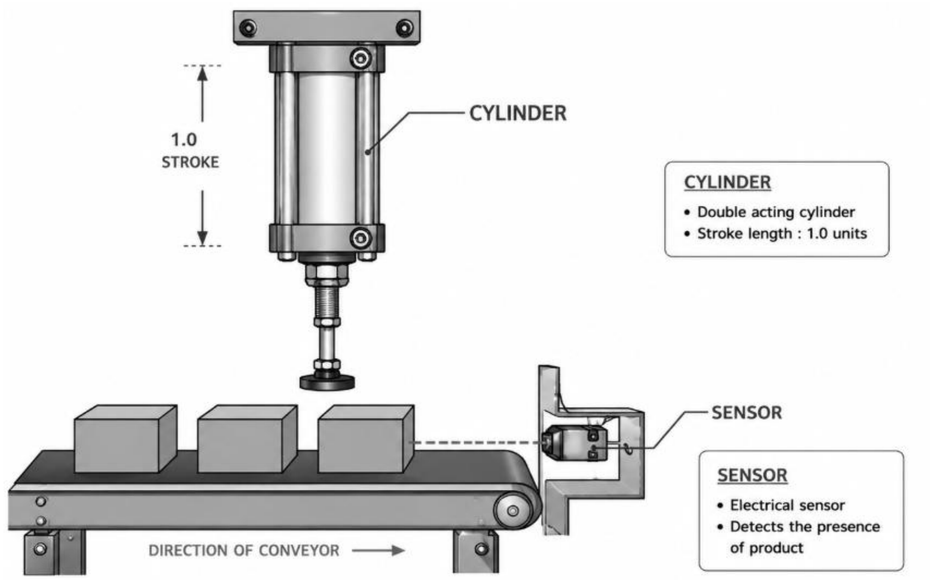
ELECTRO-PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 7B	Electro-Pneumatic Circuit for Box Lifting Station
Date:	

PROBLEM:

A transfer station removes a product from a conveyor belt. If the product is detected as present (By an electrical sensor) and if the operator presses an electrical push button, the pick-up cylinder advances. Upon release of the electrical push button, cylinder is to retract to the initial position. The transfer of the product should be happened in a controlled manner. Whereas the return stroke of the cylinder should be considerably fast. A 5/2 single solenoid spring return valve should be used.



WORKING PRINCIPLE:

ELECTRO-PNEUMATIC CIRCUIT:

INFERENCE:

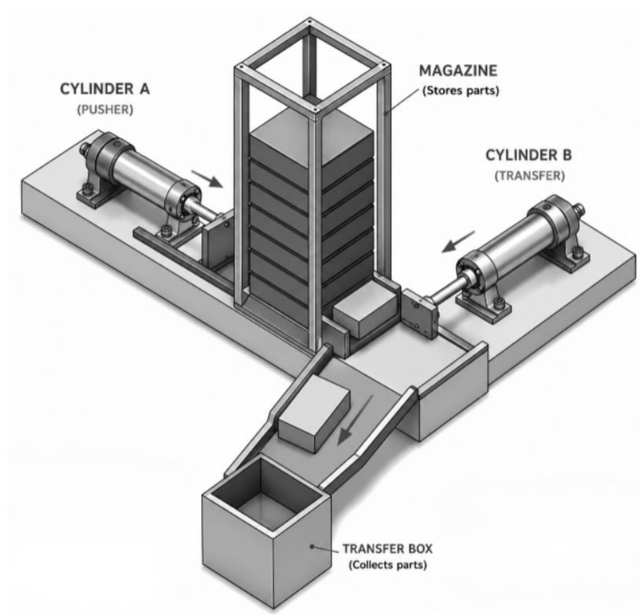
EXERCISE 8

Design, Simulation and Testing of Electro-Pneumatic Circuits with Multiple Actuators with and without Relay for Various Industrial Applications

Exp. No.: 8A	**Electro-Pneumatic Circuit Design for Transfer Station**
Date:	

PROBLEM:

Two cylinders are used to transfer parts from a magazine onto a chute. When a pushbutton is pressed, cylinder (A) extends, pushing the part from the magazine and positions it in preparation for transfer by cylinder (B) onto the out-feed chute. Once the part is transferred, the cylinder (A) retracts, followed by the cylinder (B).



Sequence code: A1 B1 A0 B0

Logic Equation: A1 = a0. Start; B1 = a1; A0 = b1; B0 = a0

WORKING PRINCIPLE:

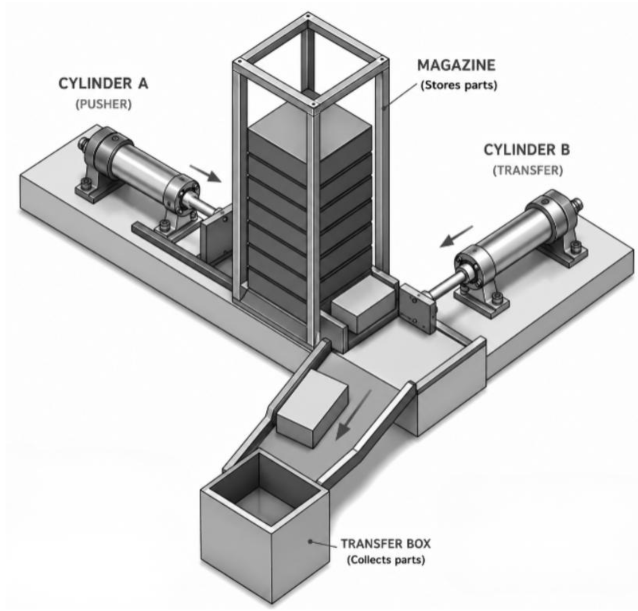
ELECTRO-PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 8B	**Electro-Pneumatic Circuit Design for Transfer Station**
Date:	

PROBLEM:

Two cylinders are used to transfer parts from a magazine onto a chute. When a pushbutton is pressed, cylinder (A) extends, pushing the part from the magazine and positions it in preparation for transfer by cylinder (B) onto the out-feed chute. Once the part is transferred, the cylinder (B) retracts, followed by the cylinder (A).



Sequence code: A1 B1 B0 A0

Logic Equation: A1 = a0. Start; B1 = a1; B0 = X1; A0 = X1. b_0; X1 = b_1; X0 = a_0

WORKING PRINCIPLE:

ELECTRO-PNEUMATIC CIRCUIT:

INFERENCE:

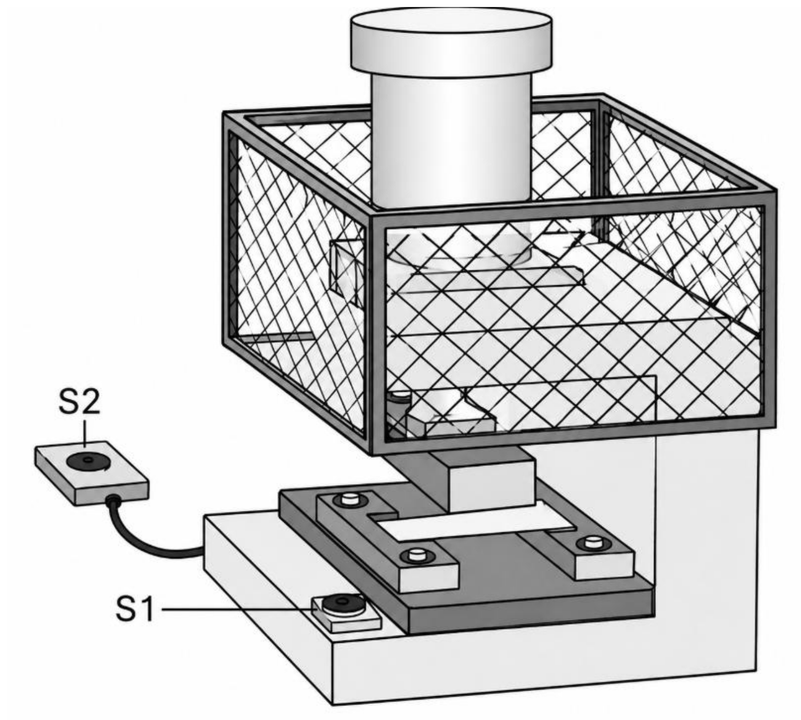
EXERCISE 9

Design and Simulation of Basic Electro-Hydraulic Circuits

Exp. No.: 9A	**Control of Hydraulic Press Using Electro-Hydraulic Circuit Along with a Speed Control**
Date:	

PROBLEM:

The ram of a hydraulic press is only to extend if the protective guard is closed which actuates switch S1 and the start button switch S2 is operated. If either of these conditions is not met, the ram is to retract immediately. The double acting hydraulic cylinder is to be operated by a 4/2 single solenoid valve with spring return and speed of ram during the forward stroke should be controlled.



WORKING PRINCIPLE:

ELECTRO-HYDRAULIC CIRCUIT:

INFERENCE:

Exp. No.: 9B	Electrohydraulic Circuit for Speed Control of Hydraulic Motor
Date:	

PROBLEM:

Design a hydraulic circuit to operate an industrial winch using a hydraulic motor. The hydraulic motor is used to move an industrial winch using a rope and rope winding mechanism. The upward and downward motion is due to the clockwise and anticlockwise movement of the motor along with the rope mechanism. The speed of the industrial winch should be controlled against the weight of the winch during the clockwise motion of the hydraulic motor (downward movement) by using counterbalance valve. Use a 4/3-way valve with spring centred hydraulic valve and operated by solenoids from both ends. Employ a start switch to operate the winch and get confirmed from the electrical limit switch on both extreme end of the winch position.

WORKING PRINCIPLE:

ELECTRO-HYDRAULIC CIRCUIT:

INFERENCE:

Additional Questions / Notes:

Additional Questions / Notes:

AMRITA VISHWA VIDYAPEETHAM

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